

Anish Aggarwal

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EXPERIENCE

Meta

London, United Kingdom

Software Engineer

July 2024 - Current

- Architected a high-performance, multi-threaded C++ data access layer for Scribe (distributed logging), optimizing for **100 TB/s** throughput with multi-tier caching, and concurrency control.
- Designed a **PID-controlled resource handler** to manage CPU/Memory budgeting during traffic spikes; eliminated **tail-latency spikes** and prevented resource exhaustion.
- Engineered an **automated A/B testing framework** to detect CPU, Network, and Memory regressions at code-review time; integrated a feedback loop that provides AI-driven optimization suggestions, preventing high-latency deployments and accelerating the developer iteration cycle.
- Partnered with ML Researchers to analyze model data-access patterns and implemented SQL-style predicate pushdown, reducing network transfer **70% (400 GB/s)** and CPU usage by **42%**, alleviating GPU compute bottlenecks and enabling faster iteration across multiple models.

Google

Warsaw, Poland

Software Developer

May 2023 - July 2024

- Improved large-scale vector retrieval latency by integrating **ScaNN** optimizing nearest-neighbor search performance for high-dimensional data under production workloads.
- Architected the lifecycle management infrastructure for VMware Admin Cluster on Google Cloud, automating resource provisioning and scaling for high-availability Kubernetes workloads
- Expanded team capacity by conducting **10+ technical interviews** and mentoring an intern through the successful delivery of a targeted feature project.

Amazon

Bengaluru, India

Software Developer

Jan 2022 - May 2023

- Designed a resilient backup payment flow that increased payment success rates by **48%**, recovering significant revenue and saving **70k+ USD** in the US region within 2 months.
- Profiled and eliminated network overhead in payment services, reducing P99 latency by **20ms** and streamlining cross-service communication.
- Designed asynchronous retry and fault-tolerance mechanisms, mitigating **34%** of transient failures.
- Spearheaded the migration of **2,000+** dependencies across core services, resolving critical security vulnerabilities and modernizing the legacy tech stack.

Elasticsearch (Google Summer of Code)

Remote

Developer

May 2020 - Aug 2020

- Implemented a highly optimized **text-diff** utility using the Myers Difference Algorithm, achieving **41ms** execution time on complex datasets.
- Built an interactive component playground for Elastic UI, streamline UI component experimentation, improving developer productivity and iteration speed.

EDUCATION

National Institute of Technology Hamirpur

Hamirpur, India

B.Tech in Computer Science and Engineering (CGPA: 9.01/10)

2018 - 2022

Relevant Coursework: Distributed Systems, Database Design, Object Oriented Design, Network Programming, Data Structures, Algorithms.

ACHIEVEMENTS

Elastic Contributor Program: Recognized as **#1 worldwide** contributor; personally awarded by the Elastic CEO.

Open Source: Active contributor to Susi.ai (Fossasia), Slate, and Swift-Nio (Apple).

SKILLS

Languages: C++ (Expert/Daily), Python, Java, SQL (MySQL), GraphQL

Systems/Cloud: Distributed Systems, Kubernetes, Docker, Google Cloud (GCP), AWS, Memcache

Performance & Tools: Linux Systems, Performance Profiling (CPU/MEM/NET), PyTorch, Elasticsearch, Git